

with probability tending to one as $G \rightarrow \infty$. The last condition in Assumption S10 simply states that asymptotically, the proportion of switchers is strictly positive.

The following result shows that under these conditions, DID_M is asymptotically normal, and that $CI_{1-\alpha}(\delta^S)$ is asymptotically conservative.

Theorem S6 *Suppose that Assumptions 1-5, 9-10 and S9-S10 hold. Then, as $G \rightarrow \infty$,*

$$\sqrt{G} (DID_M - \delta^S) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

with σ^2 defined in (38) below. Moreover,

$$\limsup_{G \rightarrow \infty} \Pr (\delta^S \in CI_{1-\alpha}(\delta^S)) \geq 1 - \alpha.$$

Theorem S6 shows that DID_M is an asymptotically normal estimator of δ^S when the number of groups tends to infinity, provided the outcomes and treatments are independent across groups. As is usually the case for estimators constructed using independent but not identically distributed random variables (see e.g. Liu and Singh, 1995), the asymptotic variance of DID_M can only be conservatively estimated. As a result, the confidence interval we propose is asymptotically conservative.

6 Detailed literature review

We now review the 33 papers that use two-way fixed effects or closely related regressions that we found in our literature review. For each paper, we use the following presentation:

Authors (year), Title. *Where the two-way fixed effects estimator is used in the paper.*

Description of the two-way fixed effects estimator used in the paper, and how it relates to Regression 1 or 2. Assessment of whether the stable groups assumption holds in this paper. Assessment of whether the research design is sharp or fuzzy.

1. **Chandra et al. (2010), Patient Cost-Sharing and Hospitalization Offsets in the Elderly.** *First line of Tables 2 and 3.*

In the regressions in the first line of Tables 2 and 3, the outcomes (e.g. a measure of utilization for plan p in month t) are regressed on plan fixed effects, month fixed effects, and an indicator of whether plan p had increased copayments in month t (see regression equation at the bottom of page 198). This regression corresponds to Regression 1. The period analyzed runs from January 2000 to September 2003. The stable groups assumption is satisfied until January 2002, when the HMO plans also become treated. This is a sharp design.

2. **Duggan and Morton (2010), The Effect of Medicare Part D on Pharmaceutical Prices and Utilization.** *Tables 2 and 3.*

In regression Equation (1), the dependent variable is the change in the price of drug j between 2003 and 2006, the explanatory variables are the Medicare market share for drug j in 2003, and some control variables. This regression corresponds to Regression 2, with some control variables. The stable groups assumption is presumably not satisfied: it seems unlikely that there are drugs whose Medicare market share in 2003 is equal to 0. This is a sharp design.

3. **Aizer (2010), The Gender Wage Gap and Domestic Violence.** *Table 2.*

In regression Equation (2), the dependent variable is the log of female assaults among females of race r in county c and year t , and the explanatory variables are race, year, county, race $\times$ year, race $\times$ county, and county $\times$ year fixed effects, as well as the gender wage gap in county c , year t , and race r , and some control variables. This regression is a “three-way fixed effects” version of Regression 1, with some control variables. The stable groups assumption is presumably satisfied: it seems likely that between each pair of consecutive years, there are counties where the gender wage gap does not change. This is a fuzzy design: the treatment of interest is the gender wage gap in a couple (see the bargaining model in Appendix 1), which varies within (year, county) cells.

4. **Algan and Cahuc (2010), Inherited Trust and Growth.** *Figure 4.*

Figure 4 presents a regression of changes in income per capita from 1935 to 2000 on changes in inherited trust over the same period and a constant. This regression corresponds to Regression 2. The stable groups assumption is satisfied: there are countries where inherited trust does not change from 1935 to 2000. This is a sharp design.

5. **Ellul et al. (2010), Inheritance Law and Investment in Family Firms.** *Table 7.*

In the regressions presented in Table 7, the dependent variable is the capital expenditure of firm j in year t , and the explanatory variables are firm fixed effects, an indicator for whether year t is a succession period for firm j , some controls, and three treatment variables: the interaction of the succession indicator with the level of investor protection in the country where firm j is located, the interaction of the succession indicator with the level of inheritance laws permissiveness in the country where firm j is located, and the interaction of the succession indicator with the level of inheritance laws permissiveness and the level of investor protection in the country where firm j is located. This regression is similar to Regression 1 with controls, except that it has three treatment variables. The stable groups assumption is presumably not satisfied: for instance, it seems unlikely that there are countries with no investor protection at all. This is a sharp design.

6. **Bustos (2011), Trade Liberalization, Exports, and Technology Upgrading: Ev-**

idence on the Impact of MERCOSUR on Argentinean Firms. *Tables 3 to 12.*

In regression Equation (11), the dependent variable is the change in exporting status of firm i in sector j between 1992 and 1996, and the explanatory variables are the change in trade tariffs in Brasil for products in sector j over the same period, and some control variables. This regression corresponds to Regression 2, with some controls. The stable groups assumption is presumably satisfied: it seems likely that there are sectors where trade tariffs in Brasil did not change between 1992 and 1996. This is a sharp design.

7. **Anderson and Sallee (2011), Using Loopholes to Reveal the Marginal Cost of Regulation: The Case of Fuel-Economy Standards.** *Table 5 Column 2.*

In the regression in Table 5 Column (2), the dependent variable is an indicator for whether a car sold is a flexible fuel vehicle, and the explanatory variables are state and month fixed effects, the percent of gas stations that have ethanol fuel in each month $\times$ state, and some controls. This regression corresponds to Regression 1. The stable groups assumption is presumably satisfied: it seems likely that between each pair of consecutive months, there are states where the percent ethanol availability does not change. This a fuzzy design: the treatment of interest is whether a car buyer has access to ethanol fuel, which varies within (month,state) cells.

8. **Bagwell and Staiger (2011), What Do Trade Negotiators Negotiate About? Empirical Evidence from the World Trade Organization.** *Table 3, OLS columns.*

In regression equations (15a) and (15b), the dependent variable is the ad valorem tariff level bound by country c on product g , while the explanatory variables are country and product fixed effects, and two treatment variables which vary at the country $\times$ product level. These regressions are similar to Regression 1, except that they have two treatment variables. The stable groups assumption is not applicable here, as none of the two sets of fixed effects included in the regression correspond to an ordered variable. This is a sharp design.

9. **Zhang and Zhu (2011), Group Size and Incentives to Contribute: A Natural Experiment at Chinese Wikipedia.** *Tables 3 and 4, Columns 4-6.*

In the regression in, say, Table 3 Column (4), the dependent variable is the total number of contributions to Wikipedia by individual i at period t , regressed on individual fixed effects, an indicator for whether period t is after the Wikipedia block, the interaction of this indicator and a measure of social participation by individual i , and some controls. This regression corresponds to Regression 1 with some controls. The stable groups assumption is satisfied: there are individuals with a social participation measure equal to 0. This is a sharp design.

10. **Hotz and Xiao (2011), The Impact of Regulations on the Supply and Quality**

of Care in Child Care Markets. *Table 7, Columns 4 and 5.*

In Regression Equation (1), the dependent variable is the outcome for market m in state s and year t , and the explanatory variables are state and year fixed effects, various measures of regulations in state s in year t , and some controls. This regression corresponds to Regression 1 with several treatment variables and with some controls. The stable groups assumption is presumably satisfied: between each pair of consecutive years, it is likely that there are states whose regulations do not change. This is a sharp design.

11. **Mian and Sufi (2011), House Prices, Home Equity-Based Borrowing, and the US Household Leverage Crisis.** *Tables 2 and 3.*

In Regression Equation (1), the dependent variable is the change in homeowner leverage from 2002 to 2006 for individual i living in zip code z in MSA m , and the dependent variable is the change in the house price for that individual, instrumented by MSA-level housing supply elasticity. This regression is the 2SLS version of Regression 2, with some controls. The stable groups assumption is presumably not satisfied: it is unlikely that some MSAs have an housing supply elasticity equal to 0. This is a sharp design.

12. **Wang (2011), State Misallocation and Housing Prices: Theory and Evidence from China.** *Table 5, Panel A.*

In regression Equation (15), the dependent variable is the quantity of housing services in household i 's residence in year t , while the explanatory variables are an indicator for period t being after the reform, a measure of mismatch in household i , the interaction of the measure of mismatch and the time indicator, and some controls. This regression is similar to Regression 1 with some controls, except that it has a measure of mismatch in household i instead of household fixed effects. The stable groups assumption is presumably satisfied: it is likely that some households have a mismatch equal to 0. This is a sharp design.

13. **Duranton and Turner (2011), The Fundamental Law of Road Congestion: Evidence from US Cities.** *Table 5.*

In the regressions presented in, say, the first column of Table 5, the dependent variable is the change in vehicle kilometers traveled in MSA s between decades t and $t-1$, and the explanatory variables are the change in kilometers of roads in MSA s between decades t and $t-1$, and decade effects. This regression corresponds to Regression 2. The stable groups assumption is presumably satisfied: it is likely that between each pair of consecutive decades, there are some MSAs where the kilometers of roads do not change. This is a sharp design.

14. **Acemoglu et al. (2011), The Consequences of Radical Reform: The French Revolution.** *Table 3.*